World Cup Prediction Model

To-Do & Next Steps

Part A — Operate the current pipeline (zero-scrape path)

1. Get the data

- martj42 **results.csv** — done.
- Download the salimt **football-datasets** dump. You need three files: **player_market_values**, **player_national_team_performances**, and **player_profiles**.

2. Run the baseline first

- Run **wc_walkforward.py** on results.csv. Confirm the Elo + form model beats the Elo-only baseline on pooled log loss and wins most folds. This is your reference point; if market value doesn't improve on it, it isn't earning its place.

3. Build the market-value feature (offline, no network)

- Load: **load_market_values**, **load_national_pool**, **load_player_profiles**.
- Resolve names: **resolve_team_names_offline(pool, profiles, market_values=mvals)** — citizenship-based, market-value-weighted so it picks senior men's teams over youth/women's sides.
- Build value: **build_country_value_series(..., squad_size=26)** — sums the top-N most valuable active players (a squad proxy), not the whole capped pool.
- Join: **align_names** then **attach_market_value(df, cs)** (leakage-safe as-of join).
- Apply **competitive_only(df)** AFTER build_dataset — drops friendlies from modelling while keeping them in the Elo/form history.
- Extend **wc.FEATURES** with mv_log_diff and mv_missing; point make_model at **make_model_with_imputer**; re-run the walk-forward.

4. Read results honestly

- Judge by the per-fold **edge** consistency (folds beating Elo: N/M), not the average. A noise feature can help on average while hurting many folds.

5. Sanity checks & tuning

- Print the resolved team map and eyeball ~20 major nations; watch heavily naturalized squads (Qatar, etc.), which are the likeliest mis-resolves — hard-override them.
- Check the **mv_missing** rate and the "% dropped" note. High missingness for a nation you care about means a name mismatch (extend NAME_ALIAS) or value-coverage gap.
- Tune **liveness_months** (default 18) if pools look stale or thin.
- Compare **squad_size=26** vs **None** to confirm the squad proxy actually helps.

Part B — Push past the model ceiling

The ceiling is for *unordered 1X2 classification from this dataset*. The way past it is a richer target, a new information source, or optimizing the betting objective — not more features in the same frame. Roughly in order of leverage:

1. Change the target: a goals model (Dixon-Coles / bivariate Poisson) — top pick

- Model expected goals per side, then derive the full scoreline matrix. May shave 1X2 log loss by fixing the draw-class structure (where your residual concentrates), and — the real payoff — unlocks totals, Asian handicaps, and correct-score markets where bookmaker lines are softer than the heavily-bet 1X2. Same data.

2. Model the market, not the match

- For betting, lower outcome log loss is not profit. Devig odds properly (multiplicative / Shin), make **Closing Line Value** the north-star metric (it predicts long-run profit far better than backtested ROI), and aim the model at the *residual* — where and why you disagree with the opening line — rather than predicting outcomes from scratch.

3. New information: availability

- The only feature class likely to beat the market: probable XI / key-player absences, applied as an adjustment to squad value (drop absent players from the top-N). A cheaper cousin is recent club form/minutes of each nation's key players — a sharpness signal Elo can't see. Hard historically; gettable at tournament time.

Secondary (same data, modest gains)

- Dynamic Bayesian ratings (Glicko or a Kalman state-space rating) that model rating *uncertainty* — matters for nations that play rarely, where fixed-K Elo is overconfident.
- Market blend: a learned weighted average of model and devigged market probabilities, for any market you are not betting into.

Avoid

- More engineered features on this data (proven tapped out).
- Bigger or deep models — you are signal-limited, not capacity-limited; they will overfit.

Calibration: international 1X2 on major matches is among the most efficient betting markets in sport. A realistic aim is matching the closing line and finding soft spots in secondary markets and lower-profile games — which collides with Europe-centric squad coverage, since the lesser games are exactly where the squad data thins out.